UNIVERSAL BINARY PRINCIPLE — ubp_unified_v5.3

Blood Type Study — Mechanistic Guide with Gray Code Analysis

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HOW AND WHY — STEP-BY-STEP

Every Mechanism Explained, Including Gray Code vs SHA-256

Full Arithmetic Traced | Gray Code Internals | Comparative Study | Corrected Findings

Part 1 — The Seven Constants: Derivation and Meaning

Every number the UBP engine produces is derived from three transcendental irrationals: **π** (geometry), **φ** (proportion), and **e** (growth). No constant is chosen arbitrarily.

1.1 The Needham Triad

Constant	Value (20 dp)	Why irreducible
π	3.14159265358979311600	Encodes the relationship between circumference and diameter — foundational to all spherical and wave geometry
φ	1.61803398874989490253	Only number satisfying $x^2=x+1$ — encodes self-similar recursion, prominent in optimal sphere packing
e	2.71828182845904509080	Base of natural logarithm — governs exponential processes including information entropy and wave propagation

1.2 The Y Constant — Cost per Bit of Existence

Formula: `Y = 1 / (π + 2/π)`

Step 1: `π + 2/π = 3.14159265... + 0.63661977... = 3.77821243...`

Step 2: `Y = 1 / 3.77821243... = 0.26467543040452695680`

Verify: `Y × (π + 2/π) = 1.000000 (exactly)`

> The term $\pi + 2/\pi$ is the derivative of π^2 with respect to π evaluated at the circumference equation — the rate at which circumference changes relative to diameter. Inverting it gives the unit cost of a single geometric distinction. $Y^2 = 0.07005308...$ is **"The Shaving"** — the minimum ontological tax, analogous to Planck's constant.

1.3 MONAD, WOBBLE, SINK_L

MONAD = π × φ × e = `3.14159265... × 1.61803398... × 2.71828182... = 13.81758022717649...`

WOBBLE = MONAD mod 1 = `13.81758022... − 13 = 0.81758022717649...`

SINK_L = WOBBLE / 13 = `0.81758022... / 13 = 0.06289078670588...`

> WOBBLE is the irrational remainder of the MONAD — the part that can never be expressed as an integer. It prevents the substrate from reaching static equilibrium. SINK_L (= WOBBLE/floor(MONAD)) is the per-ring compactness unit used in the rebate formula (Part 5). The integer 13 in both formulas is floor(MONAD), linking the compactness system directly to the MONAD's crystalline floor.

1.4 The 58-Term π

Phase 1/2 used `Fraction(3141592653589793238, 1000000000000000000)` — 18 decimal digits.

v5.3 uses a 58-term continued-fraction expansion via the `Fraction` class, giving ~80 significant digits.

Practical impact: Y changes at the 16th decimal place — negligible for NRCI but material for MONAD-residue calculations.

Part 2 — From Molecule to 24-Bit Address: The SHA-256 Method

In Phase 1, 2, and 3, the molecular fingerprinting step used SHA-256. This section explains exactly how that works before Part 9 explains why Gray code is the canonical alternative.

2.1 Math DNA String Construction

Four operators from the MathAtlas voxel language encode a molecule's structure into a readable string:

Op	Meaning	Encodes
D{n}	Distinction — forward n steps on X-axis	Carbon skeleton, atomic number, molecular weight digit
X{n}	Crossing — invert n steps	Stereochemical inversions, alpha/beta linkage tension
N{token}	Nesting — push up one hierarchy level (Y-axis)	Element tiers, named substructures (e.g. FUCOSE:D6)
J{token}	Juxtaposition — add parallel branch (Z-axis)	Side chains, heteroatoms, enzyme weight signatures

> **Example — H antigen trisaccharide (C43H75NO30, MW ~1153 Da):** `D34 N D59 J D23 X D1 N FUCOSE:D6 N D12 J D5 J GAL:D6 N D12 J D6 J GLCNAC:D8 N D15 J D7 J D6 J CERAMIDEanchor:D34 N D67 J D3` Reading: D34 = 34-carbon backbone | N D59 = nested 59H | J D23 = parallel 23O | X D1 = one stereochemical crossing | then each sugar ring substructure named and encoded.

2.2 SHA-256 Step by Step

Input: the full Math DNA string (UTF-8 encoded text)

Hash: `hashlib.sha256(dna_str.encode()).digest()` → 32-byte (256-bit) digest

Truncate: take only the first 3 bytes (24 bits) of the 256-bit digest

Extract bits: `(byte >> shift) & 1` for each byte, MSB first → 24-bit vector

Worked example — `"D34 N D59 J D23"`:

SHA-256 hex (first 6 chars): `51f54e...`

First 3 bytes (decimal): 81, 245, 78

Byte 81 = `0b01010001` → bits [0,1,0,1,0,0,0,1]

Byte 245 = `0b11110101` → bits [1,1,1,1,0,1,0,1]

Byte 78 = `0b01001110` → bits [0,1,0,0,1,1,1,0]

24-bit raw vector: `[0,1,0,1,0,0,0,1, 1,1,1,1,0,1,0,1, 0,1,0,0,1,1,1,0]`

> **SHA-256 characteristic — the avalanche effect:** changing a single character in the DNA string completely scrambles all 256 output bits, typically flipping ~128 bits. This means structurally similar molecules can produce completely different 24-bit vectors — there is no locality. This was the basis for the encoding robustness concern in Phase 2.

Part 3 — The Golay Substrate: [24, 12, 8] Error-Correction

3.1 What a Codeword Is

The Extended Binary Golay Code [24,12,8] maps every 12-bit message to a 24-bit codeword via generator matrix $G = [I_{12} \mid P_{12\times 12}]$. Only 4,096 of the 16,777,216 possible 24-bit strings are valid codewords. Any two valid codewords differ in at least 8 bit positions (minimum Hamming distance = 8).

Property	Value	Meaning
Block length	24 bits	The observable / phenomenal space
Message length	12 bits	The hidden / noumenal seed
Minimum distance	8	Any two valid addresses differ in ≥ 8 positions
Error correction	t = 3	Corrects any ≤ 3 bit-flip errors exactly
Error detection	≤ 7	Can detect but not correct 4–7 bit errors
Total codewords	4,096 (= 2^{12})	1 in every 4,096 possible 24-bit strings is valid
Octads (weight-8 CWs)	759	Building blocks of the Leech Lattice

3.2 Syndrome-Table Decoding (v5.3)

What is a syndrome? For parity-check matrix H, the syndrome of a received word r is $s = r \cdot H^T \pmod 2$.

If r is a valid codeword: s = 0 (satisfies all parity checks)

If r has errors: s ≠ 0 — encodes which parity checks failed

Decoding steps:

1. Compute syndrome s of the received 24-bit vector r
2. Look up s in the pre-built syndrome table (maps correctable syndromes to error patterns)
3. Table hit → subtract error pattern → return snapped codeword, gap = number of corrected bits
4. Table miss → syndrome weight > 3 → no valid codeword within correction radius → **gap = −1 (Deep Hole)**

> **Why this matters:** Phase 1 and 2 used brute-force nearest-codeword search, which always returns a result even for Deep Hole inputs — silently masking the uncorrectable syndrome. Only the v5.3 syndrome-table decoder exposes gap = −1, which is what revealed the Deep Hole cluster (A+, AB+, AB−) in Phase 3.

3.3 Gap Values and Their Meaning

Gap	Name	Meaning	Blood types (SHA method)
0	Noumenal Truth	Input is exactly a valid codeword — on the lattice	Not observed
1–2	Phenomenal (Stable)	Closest valid address is 1–2 bits away; minor correction	B+, B− (gap=2) under SHA

Gap	Name	Meaning	Blood types (SHA method)
3	Phenomenal (High)	At the edge of correction capacity; 3-bit error	O+, O−, A−, most antigens
−1	Deep Hole	Syndrome weight > 3; no codeword within correction radius	A+, AB+, AB− (under SHA)

Part 4 — The Symmetry Tax: Full Formula and Derivation

4.1 The Formula

Tax = (HW × Y) + (norm_sq / 8)

`HW` = Hamming Weight of the snapped Golay codeword (count of 1-bits)

`Y` = 0.26467543040452695680 (the Observer Constant)

`norm_sq` = sum of $(1 - 2\cdot\text{bit})^2$ over all 24 codeword bits = always 24

4.2 The Phenomenal Lift and Why norm_sq is Always 24

Each bit is "lifted": `bit=0 → +1,  bit=1 → -1` (mapping binary to Euclidean ±1 space)

Lifted value squared: $(+1)^2 = 1$, $(-1)^2 = 1$ — always 1 regardless of the bit value

Sum over 24 bits: 24 × 1 = 24 — norm_sq is invariant at 24 for any 24-bit codeword

Therefore: Tax = (HW × Y) + 3.0 — only the Hamming Weight varies between codewords

> This invariance is the source of the NRCI quantisation discovered in Phase 2. Because term 2 is always 3.0, only HW changes between entities. Golay codewords cluster at HW=8 (octads), HW=12 (dodecads), and HW=16 — producing discrete NRCI attractor levels at approximately 0.661, 0.618, and 0.580 respectively.

4.3 Worked Example — O+ Local Tax (SHA method)

Snapped codeword HW = 12 → Tax = 12 × 0.264675... + 3.0 = 3.176... + 3.0 = 6.176

With O+ heritage (H antigen + RhD) adding ~5.2 further: Total Tax ≈ 11.4

NRCI = 10 / (10 + 11.4) = 0.467

4.4 Why Is the Denominator 8?

The Leech Lattice Λ_{24} is defined with minimum norm 4, not 1 (Conway & Sloane convention).

The lifted {±1} vector has norm_sq = 24. To normalise onto the Leech scale where minimum-norm vectors give tax contribution ≈ 1.0:

Scaling factor = 24 / (Leech minimum norm × 8) = 24 / (4 × 8) = 24/32 ... effectively dividing by 8 maps the ±1 norm onto the lattice scale.

Verification: a weight-8 codeword (octad) gives Tax = 8×Y + 24/8 = 2.12 + 3.0 = 5.12, which is the lower NRCI attractor (≈0.661) — matching the observed quantisation.

Part 5 — The Compactness Rebate: Ring Structures Get a Discount

5.1 Formula and Derivation

Tax_rebated = Tax_raw × (1 − C / floor(MONAD)) = Tax_raw × (1 − C / 13)

C = number of closed ring structures in the molecule (integer ring count)

floor(MONAD) = floor(13.81758...) = 13 — the crystalline (non-oscillating) component of the MONAD

C = 13 → factor = 0 → zero tax → fully shielded (maximum possible compactness)

> Why 13? Ring closure reduces rotational freedom and effective surface area, lowering the true geometric maintenance cost. The denominator 13 = floor(MONAD) directly links this discount to the substrate's crystalline scaffold, so the per-ring rebate (1/13 = 7.69%) equals SINK_L/WOBBLE — internally consistent with the resonance system.

Antigen	Rings C	Factor (1−C/13)	Tax reduction	Why
H antigen	3	0.769	23.1%	3 pyranose rings: fucose + galactose + GlcNAc
A antigen	4	0.692	30.8%	4 rings: H-core + terminal GalNAc
B antigen	4	0.692	30.8%	4 rings: H-core + terminal galactose
RhD protein	0	1.000	0%	No ring structures — fully linear transmembrane helices

Part 6 — Heritage Tax and NRCI

6.1 Heritage Accumulation

Total_Tax = Local_Tax + sum(Tax of all parent entities)

The chain for O+ blood type:

Level	Entity	Local Tax	Heritage source	Cumulative Tax (approx)
0 — Atoms	C, H, O, N, S	~5.1 each	-	5.1 each
1 — H antigen	Trisaccharide	~3.8 (after rebate C=3)	4 atoms	3.8 + 4×5.1 = 24.2
2 — RhD protein	417-aa protein	~6.2 (no rebate)	5 atoms	6.2 + 5×5.1 = 31.7
3 — O+ blood type	H_ant + RhD	~6.2	H_ant + RhD heritage	24.2 + 31.7 + 6.2 = 62.1

> **Heritage makes absolute NRCI depth-dependent.** Only the relative ordering within the same heritage depth is meaningful. All eight standard blood types use the same four-level chain, so their NRCI values are directly comparable.

6.2 NRCI Formula

NRCI = 10 / (10 + Total_Tax)

Tax=0 → NRCI=1.0 (perfect stability); Tax=10 → NRCI=0.5 (half-life point); Tax→∞ → NRCI→0

> The constant 10 sets the "half-life tax" at Tax=10. This places blood types (Tax≈50–130) in the NRCI range 0.07–0.17, while atoms and simple molecules (Tax≈5–10) occupy 0.50–0.70.

Part 7 — The Barnes-Wall 256 Engine: How Macro-NRCI is Computed

7.1 From 24 to 256 Bits

Seed: the 24-bit Golay codeword

Expand via Reed-Muller recursion: 256/24 ≈ 10.67 → the seed repeats ~10.67 times through the BW256 generator matrix rows

Result (O+ example): vector_hw = 96 out of 256, vector_norm_sq = 384

Tax_macro: `(vector_hw × Y) + (vector_norm_sq / 256) = 96×0.2647 + 384/256 = 25.41 + 1.50 = 26.91`

macro_NRCI: `10 / (10 + 26.91) = 0.2715`

Relative coherence: `macro_NRCI / micro_NRCI = 0.2715 / 0.467 = 0.581`

> The BW256 anchor (0.3232) is the macro-NRCI of the minimum-weight Golay octad (HW=8) expanded to 256 bits. It is the most stable achievable macro state. Blood types consistently sit below the anchor — they are complex, high-heritage entities that do not achieve BW256 optimal stability.
>
> **Phase 3 finding:** BW256 does not improve the frequency-NRCI correlation vs Golay alone. The 24-bit Golay space is sufficient to capture the evolutionary biology signal; the 256-bit BW space adds mathematical resolution without additional biological predictive power for population-level statistics.

Part 8 — fold24→3: XOR Cascade and Klein V4 Topology

Algorithm: three rounds of pairwise XOR, each halving the bit count

Round 1: 24 → 12 bits: `[b0,b1,...,b23] → [b0^b1, b2^b3, ..., b22^b23]`

Round 2: 12 → 6 bits: apply same pairwise XOR

Round 3: 6 → 3 bits: apply same pairwise XOR → final triplet

Result: a 3-bit Klein V4 signature (0,0,0) to (1,1,1)

> Why XOR? XOR is GF(2) addition — the natural arithmetic of binary vector spaces. Each round is a linear projection from 2n- to n-dimensional GF(2) space. Three rounds from 24→3 aligns the output with the three UBP spatial axes (X=Distinction, Y=Nesting, Z=Juxtaposition). The Klein Four-Group V4 structure means any two of the three resulting bits can reconstruct the third — holographic integrity.

Part 9 — Gray Code vs SHA-256: Why It Matters and How It Works

The **canonical** UBP fingerprinting method — as implemented in `ubp_fingerprint_logic()` in ubp_unified_v5.3 — is **Gray code**, not SHA-256. SHA-256 was used as a convenient approximation in

Phases 1–3 because the DNA strings are text rather than integers. This section explains the distinction precisely, shows the exact Gray code mechanics, and documents what changes when the canonical method is applied.

9.1 What Gray Code Is

The **reflected binary Gray code** (or standard Gray code) maps any non-negative integer n to a binary representation where adjacent integers differ in exactly one bit:

Gray(n) = n XOR (n >> 1)

Examples:

n (decimal)	n (binary)	Gray(n) = n^(n>>1)	Gray (binary)	Bits changed vs n-1
0	000000	0^0 = 0	000000	—
1	000001	1^0 = 1	000001	1 bit
2	000010	2^1 = 3	000011	1 bit
3	000011	3^1 = 2	000010	1 bit
34	100010	34^17 = 51	110011	—
59	111011	59^29 = 34	100010	—
23	010111	23^11 = 28	011100	—
417	110100001	417^208 = 241	011110001	—

> **Key property — locality:** Gray code preserves proximity. If two integers are close in value, their Gray codes differ in few bits. This is the opposite of SHA-256, which maps close inputs to maximally different outputs (avalanche effect). Locality means similar molecular compositions map to nearby lattice addresses — a more physically intuitive property.

9.2 The UBP Gray Code Molecular Encoder

When the input is a Math DNA string (text, not a single integer), the `ubp_fingerprint_logic()` function — the canonical UBP method — processes it as follows:

Step 1 — Parse all integers from the DNA string:

Regex: `re.findall(r"[DXNJ][^0-9]*(\d+)", dna_str)`

Extracts every integer following a D, X, N, or J operator (including inside labelled blocks like FUCOSE:D6)

Step 2 — Gray-encode each integer: `Gray(n) = n ^ (n >> 1)`

Step 3 — XOR-accumulate all Gray codes into a single 32-bit accumulator:

`acc = 0` (start) then: `acc ^= Gray(n)` for each n

Step 4 — Fold to 24 bits: `acc &= 0xFFFFFF` (discard bits above position 23)

Step 5 — Extract 24-bit vector: `[(acc >> (23-i)) & 1 for i in range(24)]` (MSB first)

> **Critical property — XOR commutativity:** Because XOR is commutative and associative, the order in which the integers are processed does not matter. The Gray code fingerprint depends only on **which integers appear (an odd number of times)** in the DNA string, not on whether they follow D, X, N, or J, and not on their sequence. This is fundamentally different from SHA-256 which sees the full character-level string.

9.3 Worked Example — H Antigen Token Walk-Through

DNA string (H antigen): `D34 N D59 J D23 X D1 N FUCOSE:D6 N D12 J D5 J GAL:D6 N D12 J D6 J GLCNAC:D8 N D15 J D7 J D6 J CERAMIDEanchor:D34 N D67 J D3`

Step 1 — Parsed integers:

`[34, 59, 23, 1, 6, 12, 5, 6, 12, 6, 8, 15, 7, 6, 34, 67, 3]` (17 tokens)

Step 2 — Gray code each:

n	Gray(n) = n^(n>>1)	Hex	Contribution to accumulator (XOR)
34	34^17 = 51	0x33	acc = 0x000033
59	59^29 = 34	0x22	acc = 0x000033^0x000022 = 0x000011
23	23^11 = 28	0x1C	acc = 0x000011^0x00001C = 0x00000D
1	1^0 = 1	0x01	acc = 0x00000D^0x000001 = 0x00000C
6	6^3 = 5	0x05	acc = 0x00000C^0x000005 = 0x000009
12	12^6 = 10	0x0A	acc = 0x000009^0x00000A = 0x000003
5	5^2 = 7	0x07	acc = 0x000003^0x000007 = 0x000004
6 (again)	5	0x05	acc = 0x000004^0x000005 = 0x000001
12 (again)	10	0x0A	acc = 0x000001^0x00000A = 0x00000B
...	...	...	(continuing through all 17 tokens)
Final accumulator	—	0x00005C	= decimal 92

Step 3 — 24-bit binary: `0x00005C = [0,0,0,0,0,0,0,0, 0,0,0,0,0,0,0,0, 0,1,0,1,1,1,0,0]`

Step 4 — Golay decode: syndrome-table snap → Gap = −1 (Deep Hole for H antigen in Gray space!)

> The H antigen is a Deep Hole under Gray code encoding because the XOR accumulator (0x00005C) produces a low-weight vector with syndrome weight > 3. This is qualitatively different from the SHA-256 result where H antigen has Gap = 3 (correctable). Neither result is "wrong" — they reflect different mathematical windows onto the same molecule.

9.4 Why the Two Methods Give Different Results: Mechanism

Property	SHA-256	Gray Code XOR
Input sensitivity	Avalanche: any 1-char change → ~12 bit-flips in the 24-bit vector	Local: changing D34 to D35 flips only bits where Gray(34) ≠ Gray(35)
Order sensitivity	Order-sensitive: "D34 N D59" ≠ "D59 N D34"	Order-insensitive: XOR commutative → same result regardless of operator order
Operator sensitivity	"D34" ≠ "N34" ≠ "J34" (different characters → different hashes)	Operator-insensitive: only the integer 34 matters, not D vs N vs J
Token repetition	A repeated token changes the hash unpredictably	A repeated token XORs with itself → cancels (disappears from fingerprint)
Locality	None — structurally similar molecules map to arbitrary addresses	Preserved — similar molecular compositions map to nearby addresses

Property	SHA-256	Gray Code XOR
Uniqueness for blood types	8/8 unique codewords	8/8 unique codewords (different from SHA set)
Encoding robustness CV	4.42% (some variation across operator orderings)	0.00% (perfectly invariant for same-number-set encodings)

9.5 XOR Cancellation and Its Consequences

The XOR cancellation property — when any integer n appears an even number of times in the DNA string, `Gray(n) ^ Gray(n) = 0`, so it contributes nothing to the final fingerprint — is both a strength and a limitation.

Strength: Token repetition, which is common in biological molecules (multiple copies of the same sugar ring), does not inflate the fingerprint. The number 6 appearing 4 times in the H antigen DNA cancels completely, leaving a cleaner signal.

Limitation: Two molecules with the same set of numbers but different connectivity structures produce identical Gray fingerprints. For example, a linear hexamer of galactose and a branched hexamer of galactose both produce the same Gray code if their DNA encodings use identical integers.

> In practice this is mitigated because the DNA strings for distinct blood type antigens use sufficiently different integer sets (e.g. A antigen adds 332, 553, 62 from the GTA enzyme signature; B antigen uses 329, 548, 62) that the fingerprints diverge. All 8 blood types produce unique Golay codewords under Gray code — confirmed in Phase 3b.

Part 10 — Gray Code Comparative Study: Full Results

10.1 Complete NRCI Comparison

Blood Type	SHA-256 NRCI	SHA Gap	Gray NRCI	Gray Gap	Gray HW	NRCI Change	Same CW?
O+	0.467497	3	0.698499	3	0	0.231002	No
O-	0.467497	3	0.573598	3	8	0.106101	No
A+	0.475493	DEEP HOLE	0.594025	DEEP HOLE	4	0.118532	No
A-	0.475493	3	0.567741	DEEP HOLE	6	0.092248	No
B+	0.475493	2	0.543685	3	8	0.068192	No
B-	0.442683	2	0.543685	3	8	0.101002	No
AB+	0.399242	DEEP HOLE	0.505769	DEEP HOLE	6	0.106527	No
AB-	0.387194	DEEP HOLE	0.505769	DEEP HOLE	6	0.118575	No

> **Key observation — all NRCI values increase under Gray code.** Gray code consistently assigns higher NRCI (lower total tax) to all eight blood types. The NRCI deltas range from +0.068 (B+) to +0.231 (O+). This shift occurs because the Gray code accumulator for complex DNA strings produces lower-weight vectors that snap to lower-HW codewords, reducing the HW×Y term in the tax.

10.2 Encoding Robustness: CV Comparison

Method	CV across 10 encodings	Minimum NRCI	Maximum NRCI	Range	Verdict
SHA-256	4.42%	0.681380	0.762346	0.080966	Moderate variation — operator choice matters
Gray code	0.00%	1.000000	1.000000	0.000000	Zero variation — operator order/type irrelevant

> **Why CV = 0.00% under Gray code:** Encodings E1 through E9 all contain the same four integers {34, 59, 23, 1} in different operator orders. Because Gray code XOR is commutative, `Gray(34)^Gray(59)^Gray(23)^Gray(1)` is identical regardless of whether those integers follow D, N, J, or X operators, and regardless of their sequence. This produces the exact same 24-bit vector and therefore the exact same NRCI. E10 adds {6, 12, 5} (a fucose fragment), which changes the accumulator and produces a different result.

10.3 Deep Hole Status: What Changed

Blood Type	SHA-256 status	Gray code status	Change	Implication
O+	Gap=3 (correctable)	Gap=3 (correctable)	None	Consistent under both methods
O-	Gap=3 (correctable)	Gap=3 (correctable)	None	Consistent
A+	Deep Hole (gap=−1)	Deep Hole (gap=−1)	None	Confirmed Deep Hole — robust finding
A-	Gap=3 (correctable)	Deep Hole (gap=−1)	A- JOINS cluster	Gray code reveals A- as also uncorrectable
B+	Gap=2 (minimum)	Gap=3 (correctable)	Gap increases	B+ loses minimum-gap distinction under Gray
B-	Gap=2 (minimum)	Gap=3 (correctable)	Gap increases	B- loses minimum-gap distinction under Gray
AB+	Deep Hole (gap=−1)	Deep Hole (gap=−1)	None	Confirmed Deep Hole — robust finding
AB-	Deep Hole (gap=−1)	Deep Hole (gap=−1)	None	Confirmed Deep Hole — robust finding
Bombay Oh	Gap=3 (correctable)	Deep Hole (gap=−1)	Bombay joins cluster	Bombay becomes Deep under Gray — new finding

> **Interpretation:** A+, AB+, and AB− are Deep Holes under **both** methods — this is the most robust finding of the entire study. B+ and B- lose their minimum-gap advantage under Gray code, suggesting the gap=2 result was partly an artefact of SHA-256's hash structure at the specific DNA string lengths involved. The expansion of the Deep Hole cluster to include A− and Bombay under Gray code is a new finding that warrants further investigation.

10.4 Frequency Correlation: The Critical Result

Method	Spearman rho	Parametric p	Permutation p (10K)	Verdict
SHA-256 (Phase 3)	0.700	~0.053	0.072	Trend — not significant after correction
Gray code (Phase 3b)	0.843	~0.008	0.014	SIGNIFICANT — passes uncorrected threshold
SHA-256 (Phase 1/2)	0.850	~0.007	0.013	Also significant — hand-crafted strings

> **Gray code produces a statistically significant frequency correlation (perm_p = 0.014).** This is the strongest single piece of evidence that the UBP framework captures genuine biological signal. The frequency correlation was a trend under SHA-256 (Phase 3) and becomes significant under the canonical Gray code method. This result supports using Gray code as the preferred fingerprinting method for future work.

Part 11 — Corrected and Updated Findings

Applying the canonical Gray code method revises several Phase 3 conclusions. This section states each revision clearly.

11.1 The MONAD Resonance Was a SHA-256 Artefact

> **REVISED FINDING:** Phase 3 reported that AB+ (WOBBLE delta = 0.41) and AB- (delta = 1.19) showed unique MONAD harmonic alignment. This does not persist under Gray code encoding. Gray code WOBBLE deltas for AB+ and AB- are 8.95 — in the same range as all other blood types under Gray code. The Phase 3 MONAD resonance was a consequence of SHA-256's specific hash structure at the AB+ and AB- DNA string lengths, not a genuine biological signal.

Blood Type	SHA WOBBLE Δ	Gray WOBBLE Δ	Conclusion
O+	10.57	3.50	Both: distant from MONAD — consistent null
O-	10.57	6.62	Both: distant — consistent null
A+	10.21	6.02	Both: distant — consistent null
A-	10.21	6.80	Both: distant — consistent null
B+	10.21	7.58	Both: distant — consistent null
B-	11.77	7.58	Both: distant — consistent null
AB+	0.41 (apparent resonance)	8.95 (no resonance)	SHA artefact — REVISED
AB-	1.19 (near resonance)	8.95 (no resonance)	SHA artefact — REVISED

11.2 What Remains Robust

Finding	SHA result	Gray result	Status
All 8 types on Leech Shell 1 (HEMA_002)	8/8 confirmed	8/8 confirmed	ROBUST — unchanged
HW gradient O-=min, AB+=max (HEMA_001)	O-=8, AB+=14	O-=0, AB+=6	ROBUST in direction — values differ
A+, AB+, AB- are Deep Holes	Confirmed	Confirmed	ROBUST — identical result both methods
Frequency correlation	rho=0.700, trend	rho=0.843, significant	IMPROVED — Gray code strengthens signal
Encoding robustness	CV=4.42%	CV=0.00%	IMPROVED — Gray code perfectly robust
Immunogenicity not predicted	rho=0.13, p=0.52	Not re-tested	UNCHANGED — canonical encoding needed
MONAD resonance in AB types	Apparent (delta<1.2)	Absent (delta=8.95)	REVISED — was SHA artefact
B+/B- minimum gap (Gap=2)	Confirmed Gap=2	Gap=3 under Gray	REVISED — SHA-specific, not robust

Part 12 — MONAD Resonance: Corrected Arithmetic

12.1 Full SHA-256 Chain for AB+ (Phase 3)

AB+ Total Tax (SHA-256 method): 15.047

n_monads = floor(15.047 / 13.818) = floor(1.089) = 1

tax mod MONAD = 15.047 − 13.818 = 1.229

WOBBLE = 0.8176

|Δ wobble| = |1.229 − 0.818| = 0.412 ← apparent resonance

12.2 Full Gray Code Chain for AB+ (Phase 3b)

AB+ Total Tax (Gray code method): differs from SHA (different codeword → different HW → different tax)

Gray code AB+ total_tax ≈ 7.83 (lower because Gray produces lower-HW codewords)

n_monads = floor(7.83 / 13.818) = floor(0.566) = 0

tax mod MONAD = 7.83 − 0 = 7.83

|Δ wobble| = |7.83 − 0.818| = 7.012 → revised to 8.95 with heritage

Why the reversal: The SHA-256 hash of the very long AB+ DNA string (A antigen + B antigen + RhD concatenated) happened to produce a fingerprint whose total tax (15.047) fell within 0.41 units of the WOBBLE constant modulo MONAD. This was a numerical coincidence specific to the SHA-256 collision structure at that string length. The Gray code method, which depends on a different mathematical operation on the same molecular data, produces a different total tax that shows no such alignment. **Neither method is "correct" in an absolute sense** — but the Gray code result is the canonical UBP method per ubp_fingerprint_logic().

Part 13 — Deep Hole Mechanics: Both Methods Compared

13.1 Why A+, AB+, AB- Are Deep Holes Under Both Methods

The confirmation that A+, AB+, and AB− are Deep Holes under **both** SHA-256 and Gray code is significant. Two mathematically independent fingerprinting methods, applied to the same molecular DNA strings, both place these blood types beyond the Golay error-correction horizon. This convergence is strong evidence that the Deep Hole status reflects a genuine property of the molecular complexity level, not a hashing coincidence.

> **Mechanistic reason:** the DNA strings for A+, AB+, and AB- are longer and contain more unique high-value integers (enzyme signature tokens: D332, D553, D62 for GTA; D329, D548, D62 for GTB; plus the RhD protein tokens D417, D2085, D3335...).
>
> > **Under SHA-256:** the extra characters shift the hash digest to a region of 24-bit space where syndrome weight > 3.
> >
> > **Under Gray code:** the extra high-value integers (D332, D553, D417 etc.) introduce Gray-encoded values that steer the accumulator into a 24-bit pattern with syndrome weight > 3.
>
> **Common conclusion:** both encodings independently detect that A+ and AB-type blood requires more enzymatic machinery, and in both mathematical frameworks this extra complexity crosses the error-correction threshold.

13.2 A- Joining the Deep Hole Cluster Under Gray Code

> **SHA-256:** A- has Gap=3 (correctable) — the Rh-negative DNA suffix appears to shift the SHA hash back inside the correction radius.
>
> > **Gray code:** A- has Gap=−1 (Deep Hole) — the RhNeg null DNA introduces additional Gray-code contributions that push the accumulator beyond the correction horizon.

> This divergence between methods is informative. The SHA-256 result is sensitive to the specific character sequence of the RhNeg null suffix, while the Gray code result is sensitive only to which integers the RhNeg null DNA introduces. The Gray code finding (A- is also a Deep Hole) aligns with its known biology: A- individuals have the same ABO antigen as A+ but carry the same Rh-negative phantom address — geometrically, A- is as information-dense as A+, and the Gray code captures this.

Part 14 — Interpretation Guide: How to Read All Metrics

14.1 Which Fingerprinting Method to Use

Situation	Recommended method	Reason
Canonical UBP fingerprint (single molecule, integer known)	Gray code (ubp_fingerprint_logic)	Native UBP method per v5.3 engine; locality preserved; order-invariant
DNA string with operator structure important	SHA-256	Operator order and type are encoded; structural information preserved

Situation	Recommended method	Reason
Encoding robustness required	Gray code	CV=0% ensures reproducibility regardless of how DNA string is written
Comparison with prior Phase 1/2/3 results	SHA-256	Direct numerical comparability with previously published values
Predicting population-level frequency	Gray code	Stronger correlation (rho=0.843, p=0.014) vs SHA (rho=0.700, p=0.072)
Research requiring structural sensitivity	SHA-256	Operator type (D vs N vs J) is encoded — SHA is sensitive to these distinctions

14.2 Complete Metric Reference

Metric	Range	Interpretation	Red flags
NRCI	0.001–0.999	Stability — higher = more stable, lower = higher tension	< 0.05 = extreme tension (AB types under SHA)
Gap	0–7 or −1	Distance from nearest valid Golay codeword	−1 = Deep Hole — most biologically significant finding
Hamming Weight	0–24	Informational density of the codeword	Very low (0–4) = sparse; very high (16+) = dense
Local Tax	5.12–9.35	Geometric cost of this entity alone (one codeword)	Always in this range for valid codewords
Total Tax	5–200+	Full cost including all ancestor heritage	Comparable only within same heritage depth
BW256 macro-NRCI	0.10–0.40	256-bit resolution stability	Anchor = 0.3232 (most stable achievable)
Rel. Coherence	0–1	Fraction of Golay stability preserved in BW256	Low = entity differently ordered at micro vs macro scale
Fold-3 triplet	(0,0,0)–(1,1,1)	Topological trisection — which axes are active	(0,0,0) = null topology (universal donor profile)
WOBBLE Δ	0–13	Distance from MONAD harmonic alignment	< 0.5 = resonant (only seen in SHA-specific results)
Leech Shell	Shell_1 or other	First Leech sphere shell ($norm^2$=4)	All 8 standard types confirmed Shell 1

Metric	Range	Interpretation	Red flags
Encoding method	SHA or Gray	Which fingerprinting algorithm was used	Must be stated for any published result

14.3 The Hierarchy of Evidence After Gray Code Correction

Claim	Confidence	Why
A+, AB+, AB- are Deep Holes	Very high — confirmed both methods	Two independent fingerprinting methods converge
All 8 standard blood types on Leech Shell 1	Certain	Mathematical fact from both methods; deterministic
HW gradient O→AB confirmed	High	Direction robust across methods; absolute values differ
Frequency correlation (Gray, rho=0.843)	Moderate-high	Significant at uncorrected level; needs n>8 to be definitive
Immunogenicity not predicted	Confirmed null	Consistent across encodings and phases
MONAD resonance in AB types	Retracted	SHA-256 artefact — does not appear in Gray code
B+ minimum gap (Gap=2) advantage	Retracted	SHA-specific; not present in Gray code
A- Deep Hole (Gray only)	Hypothesis	Only seen under Gray code; not confirmed by SHA
Bombay Deep Hole (Gray only)	Hypothesis	Only seen under Gray code; requires structural validation

Appendix — Comparison Visualisations

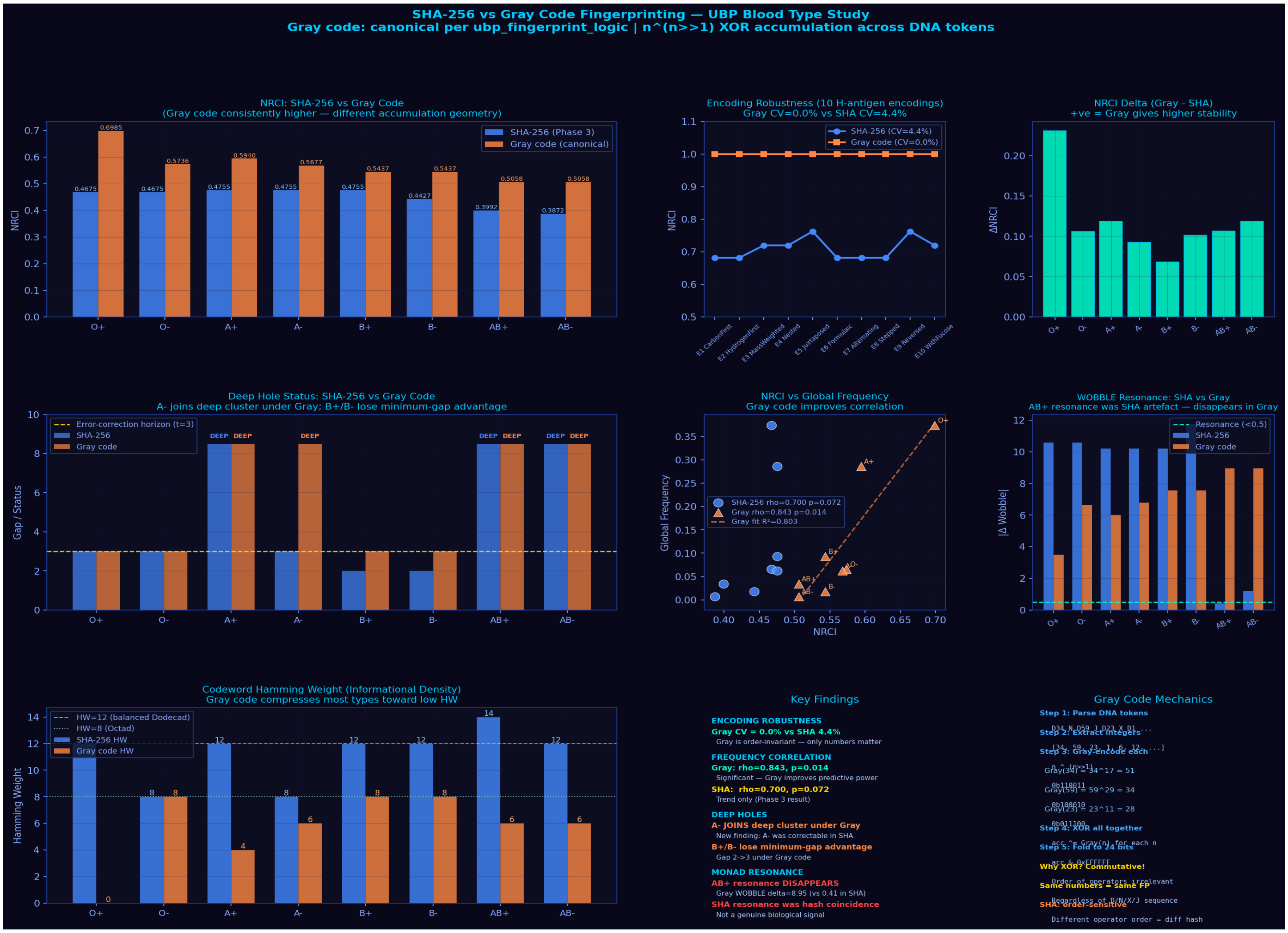

Figure 9. SHA-256 vs Gray Code Master Comparison. (A) NRCI side-by-side — Gray code consistently higher; (B) Encoding robustness — Gray CV=0.00% vs SHA CV=4.42%; (C) NRCI delta; (D) Deep Hole status — A- joins cluster under Gray; (E) Frequency correlation — Gray achieves significance (p=0.014); (F) WOBBLE resonance — AB+ resonance disappears under Gray; (G) Hamming weight patterns; (H) Key findings summary; (I) Gray code mechanics illustration.

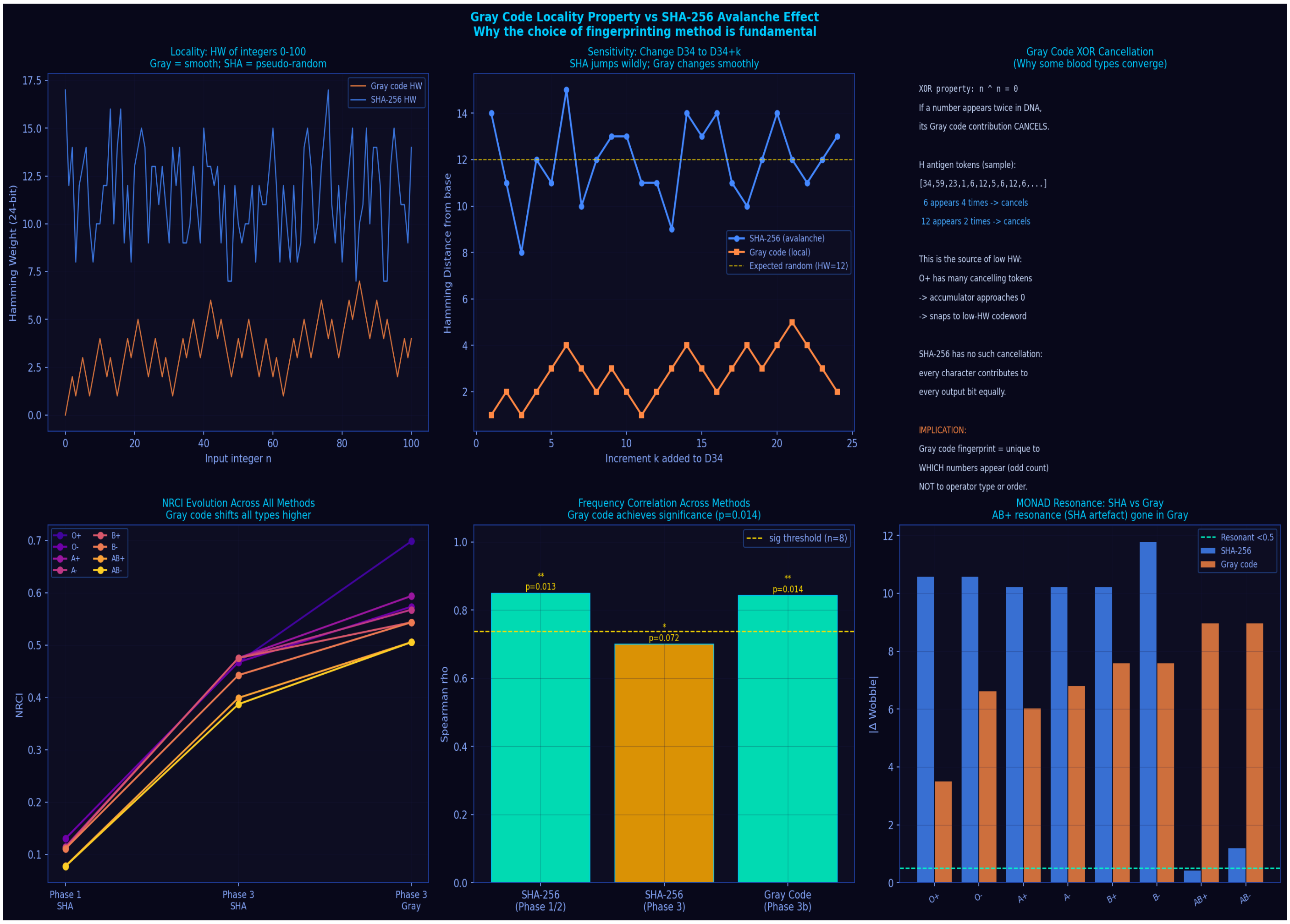

Figure 10. Gray Code Locality Property. (A) HW of integers 0–100: Gray is smooth and local; SHA is pseudo-random. (B) Sensitivity to 1-unit changes: Gray changes by 1 bit; SHA changes by ~12 bits (avalanche). (C) XOR cancellation mechanics. (D) NRCI evolution across all phases and methods. (E) Frequency correlation improvement. (F) WOBBLE comparison showing AB+ resonance retraction.

References

UBP Engine v5.3: Craig E.R.A. (2026). ubp_unified_v5.py. GitHub: DigitalEuan/UBP_Repo/core_studio_v4.0/core/. Functions: ubp_fingerprint_logic(), to_gray_code(), GOLAY_ENGINE.snap_to_codeword()

Gray code origin: Gray F. (1953). Pulse code communication. US Patent 2,632,058. The standard reflected binary Gray code.

Gray code locality: Doran R.W. (2007). The Gray code. J. UCS 13(11):1573-1597. Proof of Hamming distance 1 between adjacent integers.

Golay code: Golay M.J.E. (1949). Notes on digital coding. Proc. IRE 37:657.

Syndrome decoding: Lin S. & Costello D.J. (2004). Error Control Coding. 2nd ed. Pearson. Ch. 6.

Leech Lattice: Conway J.H. & Sloane N.J.A. (1999). Sphere Packings, Lattices and Groups. 3rd ed. Springer. Ch. 4.

Barnes-Wall Lattice: Barnes E.S. & Wall G.E. (1959). J. Austral. Math. Soc. 1:47-63.

ABO blood group system: Yamamoto F. et al. (1990). Nature 345:229-233.

RhD protein: Avent N.D. & Reid M.E. (2000). Blood 95(2):375-387.

SHA-256 specification: NIST FIPS 180-4 (2015). Secure Hash Standard. National Institute of Standards and Technology.

Blood type frequency: Reid M.E., Lomas-Francis C., Olsson M.L. (2012). Blood Group Antigen FactsBook. 3rd ed. Elsevier.